Graphics engine improvements

Airline Manager Pro · catalogue of improvements to the current engine (three.js) · 2 November 2026

How to read the tables: **Cost** is the performance cost while playing (None, Low, Medium, High; "Gain" means it makes the game faster). **Tier** is the lowest graphics setting it would be switched on at (All, Medium, High, Ultra; "Build" means it is done when the game is built, not while playing; "Dev" means a tool for making the game). Every item is written as our own code or uses MIT, zlib, Apache or CC0 material, so all are allowed in a paid game.

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Can the current engine look like Unity or Unreal?

Short answer: it can get close to Unity's look and reach a good part of Unreal's look for this kind of game, but not all of it.

Area	Unity (URP / HDRP)	Unreal Engine 5	Our engine after these improvements
Materials (metal, fabric, glass, skin)	Physically based	Physically based, very rich	Physically based, same principles: **comparable**
Lighting in rooms and the cabin	Baked lighting and probes; HDRP live options	Lumen: live bounced light everywhere	Baked lighting, light probes, ambient occlusion: **close** for rooms that don't change; live bounced light only approximated (screen-space)
Shadows	Cascaded, soft	Virtual shadow maps: sharp at any distance	Cascaded, soft, contact shadows: **close to Unity**, below Unreal at great distances
Reflections	Probes, screen-space, ray-traced (HDRP)	Lumen and ray-traced	Probes and screen-space: **good**, no ray-traced reflections
Sky, clouds, fog	Physical sky, volumetric clouds and fog (HDRP)	Sky atmosphere, volumetric clouds and fog	Same methods possible: **comparable**, cost kept down with lower resolution and upscaling

Area	Unity (URP / HDRP)	Unreal Engine 5	Our engine after these improvements
Geometry detail	Level of detail	Nanite: film-level detail with no level-of-detail work	Level of detail, instancing, normal maps: **good**, but not Nanite's density
People and faces	Good with bought assets	MetaHuman: near-photoreal	Stylised to semi-realistic: skin shading, better eyes and hair cards help a lot; **below MetaHuman**
Image quality	TAA, upscaling (FSR / DLSS)	TSR, DLSS, FSR	TAA and FSR-style upscaling: **comparable**; no DLSS (needs native code)
Ray tracing during play	HDRP, with RTX cards	Yes	**Not possible yet** in browsers; path-traced photo mode instead
Performance	Native code: fastest	Native, heavy on hardware	Browser adds some overhead; WebGPU narrows the gap. Our scenes are much smaller than a typical Unreal game, so a steady 60 fps on a mid-range PC is realistic

What decides the look more than the engine: lighting, materials, textures and the final image processing. Unreal games look the way they do mainly because of these, and because of large libraries of scanned assets. The methods are published and well documented; they can be written for our engine. The limits are what browsers allow (no hardware ray tracing, no DLSS) and the time needed to tune each scene until it looks right, not how fast the code can be written.

What can't be matched: live ray tracing, Nanite-level geometry, Lumen's fully live bounced light in big changing scenes, and MetaHuman-level faces.

1. Engine core and renderer

Improvement	What it does	Benefit	Cost	Tier
Update three.js (0.159 to the current version)	Moves to two years of newer engine work	Faster drawing, fixes, newer effects (node materials, newer post-processing) available	Gain	All
WebGPU renderer with WebGL fallback	Uses the modern graphics interface of the browser	Less work for the processor per frame, compute shaders, more objects and lights at the same speed; WebGL kept for older phones	Gain	All
Node materials (TSL)	Materials written as building blocks that work in both WebGPU and WebGL	One material system for every effect below; easier to add new looks	None	All
Compute shaders	Graphics card does calculations (particles, culling, crowds)	Thousands of particles and objects without slowing the processor	Gain	High
One shared renderer for all views	The office, cabin, court and 3D view reuse the same renderer and resources	Faster switching between screens, less memory, fewer loading pauses	Gain	All
Render graph	Each frame is planned as steps (shadows, depth, colour, effects)	Effects share data instead of drawing the scene twice; easier to switch effects per tier	Gain	All
Depth pre-pass	Draws the depth of the scene first	Expensive lighting is only calculated for pixels that are seen; needed by several effects	Low	High
Reversed depth buffer	More precise depth for far distances	No flickering where surfaces overlap (runway markings, distant terrain)	None	All
Floating origin	Keeps the camera near the centre of the coordinate system	No shaking of aircraft and cockpit far from the airport	None	All
Shader warm-up during the splash screen	Prepares all shaders while loading	No stutter the first time an effect or room is shown	None	All

Improvement	What it does	Benefit	Cost	Tier
Pipeline and shader cache	Keeps prepared shaders between sessions	Faster starts after the first one	None	All
Fixed time step for animation	Movement and effects step at an even rate	Smooth motion even when the frame rate varies	None	All
Frame pacing	Even time between frames	Feels smoother at the same frame rate	None	All
High-refresh support (120/144 Hz)	Draws as fast as the monitor allows	Very smooth camera turns on gaming monitors	Medium	High
Off-screen rendering in a worker	Drawing runs beside the menus, not inside them	Menus stay responsive while 3D is busy	Gain	All (PC)

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2. Performance and memory

Improvement	What it does	Benefit	Cost	Tier
Instancing	Draws many copies (seats, passengers, lights, binders, trees) in one call	Often the biggest speed gain; allows far more detail	Gain	All
Batched meshes	Combines different small objects into one draw	Fewer draws for offices and airports full of props	Gain	All
Merging static geometry	Joins objects that never move	Fewer draws, less processor work	Gain	All
Frustum culling tuning	Skips objects outside the view, with correct bounds	Less work every frame; no objects popping at the edges	Gain	All
Occlusion culling	Skips objects hidden behind others	Large gains in the cabin and buildings	Gain	Medium
Portal culling for rooms	Only draws rooms visible through doors and windows	Offices, court and cabin draw only what can be seen	Gain	All
Compressed textures (KTX2 / Basis)	Textures stay compressed on the graphics card	4–6 times less graphics memory, faster loading, more textures possible	Gain	All
Texture atlases and arrays	Many small textures in one	Fewer material switches, fewer draws	Gain	All
Mipmaps everywhere	Smaller texture versions for far surfaces	No shimmer, faster drawing	Gain	All
Texture streaming	Loads the sharpest texture levels only when close	High detail up close without filling memory	Gain	High
Geometry compression (meshopt / quantisation)	Smaller vertex data	Less memory and faster loading	Gain	All
Index and vertex cache optimisation	Reorders triangles for the graphics card	Faster drawing of the same shapes	Gain	All
Dynamic resolution	Lowers the drawing size slightly when the frame rate drops	Steady frame rate in heavy moments	Gain	All
Upscaling (FSR 1-style)	Draws smaller and sharpens to full size	Large speed gain at nearly the same sharpness	Gain	All
Temporal upscaling	Combines several frames to rebuild full detail	Sharper than simple upscaling at the same speed	Low	High
Variable-rate effects	Expensive effects at half or quarter resolution	Fog, ambient occlusion and clouds at a fraction of the cost	Gain	All
Shadow caching	Shadows of still objects drawn once	Big savings in rooms and at airports	Gain	All
Static reflection caching	Reflection maps updated only when needed	Reflections without a cost every frame	Gain	All
Update rate per distance	Far animations and effects update less often	Crowds and airport traffic cost less	Gain	All

Improvement	What it does	Benefit	Cost	Tier
Animation level of detail	Simpler animation for far figures	Many more people on screen	Gain	All
Object pools	Reuses objects instead of creating new ones	No stutter from memory clean-up	Gain	All
Releasing graphics memory between screens	Frees textures and models when leaving a room (data kept)	Lower memory, no slowdown after hours of play	Gain	All
Frame budget manager	Spreads heavy work (loading, baking) over several frames	No hitches when new things appear	Gain	All
Background loading in workers	Builds geometry and decodes textures off the main thread	Smooth play while airports and rooms load	Gain	All
GPU timing per effect	Measures how long each effect takes	Every effect stays within budget; tiers are set from real numbers	None	Dev

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3. Lighting

Improvement	What it does	Benefit	Cost	Tier
Physically based light units	Lights in lumens and lux; the sun in real strength	Light looks consistent and believable everywhere	None	All
Real sun position	Sun from date, time and the airport's position	Correct light for every flight and time of day	None	All
Clustered (Forward+) lighting	Splits the view into cells and lights only what each light touches	Hundreds of real lights (airport, cabin, offices) at little cost	Low	Medium
Area lights	Lights from panels and strips, not points	Soft, realistic light from ceiling panels, monitors and windows	Low	Medium
Light shapes (IES profiles)	Real lamp light patterns	Believable pools of light from spotlights and floodlights	Low	High
Light probes	Captured light at points in a room	Moving people and objects get the room's colours and brightness	Low	All
Reflection probes with box projection	Room-shaped reflections	Correct reflections in glossy floors, desks and windows inside rooms	Low	All
Image-based lighting from HDR skies	Light from a real sky picture	Natural outdoor light and reflections	Low	All
Time-of-day light blending	Smooth light changes from dawn to night	Living scenes over a long flight	Low	All
Light flicker and warm-up	Fluorescent and LED behaviour	Small realism touches in offices and hangars	None	High
Emissive surfaces that light the room	Glowing screens and signs add light	Monitors and signs feel part of the scene	Low	High
Cabin light modes	Boarding, cruise, night and landing lighting	Matches real airline cabin lighting	None	All
Exposure adaptation (eye adaptation)	Brightness adjusts like the eye when moving between dark and bright	Natural feel when entering the cabin from the apron or looking outside	Low	Medium
Light leaks control	Stops light passing through walls	No glowing seams in rooms	None	All

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4. Shadows

Improvement	What it does	Benefit	Cost	Tier
Cascaded shadow maps	Several shadow maps from near to far	Sharp shadows up close, still covering a whole airport	Medium	Medium
Soft shadows (PCSS)	Shadows softer the further from the object	Realistic shadows of aircraft and furniture	Medium	High
Contact shadows	Short screen-space shadows	Feet, cups and seat legs touch the ground properly	Low	Medium
Shadow stabilisation	Shadows don't crawl when the camera moves	No shimmering shadow edges	None	All
Point and spot light shadows	Shadows from lamps, not only the sun	Lamps on desks and hangar lights cast real shadows	Medium	High
Baked shadows for still objects	Shadows stored in the lighting	Detailed room shadows for free	None	All
Cloud shadows	Clouds darken the ground below	Moving patches of light over the islands	Low	High
Transparent shadows	Coloured and partial shadows through glass and fabric	Correct light through cabin windows and curtains	Low	High
Shadow level of detail	Far objects use simpler shadow shapes	Same look, lower cost	Gain	All

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5. Global illumination and ambient occlusion

Improvement	What it does	Benefit	Cost	Tier
Ambient occlusion (GTAO)	Darkens corners and contact areas	Objects sit in the scene; one of the biggest realism gains	Low–Medium	Medium
Baked lightmaps	Bounced light calculated once and stored	Photo-like soft light in offices, court and cabin at no play cost	None	Build
Path-traced baking	Lightmaps calculated with full light paths when the game is built	The most realistic still lighting possible	None	Build
Baked ambient occlusion in models	Contact darkening stored with each model	Detail on phones too, where live effects are too costly	None	All
Irradiance volumes	A grid of light samples through a space	Moving people pick up bounced light from walls and floors	Low	Medium
Screen-space global illumination	Approximates live bounced colour on screen	Colour bleeding (red seats tint the aisle), more depth	Medium–High	Ultra
Probe-based live global illumination (DDGI-style)	Updates light probes over time	Light that changes with time of day in the cabin and hangars	High	Ultra
Bent normals	Better direction of ambient light	Softer, more correct shading in folds and corners	Low	High
Specular occlusion	Removes reflections inside crevices	No unrealistic shine in dark gaps	Low	High
Sky visibility	Areas that see less sky get less sky light	Natural darkening under wings and in hangars	Low	Medium

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6. Materials and shading

Improvement	What it does	Benefit	Cost	Tier
All materials physically based	Replace the remaining simple materials	Consistent, realistic look in every scene	Low	All
Clear coat	A glossy layer over paint	Aircraft paint and polished desks look real	Low	Medium
Sheen	Soft edge shine of cloth	Seat fabric, carpets and uniforms look like fabric	Low	Medium
Anisotropic reflections	Stretched highlights	Brushed metal, engine cowlings, hair	Low	High
Iridescence	Rainbow sheen	Fuel on wet tarmac, coated glass	Low	High
Transmission and thickness	Light through glass and thin materials	Real windows, cups, curtains and lamp shades	Medium	High
Refraction	Bends the view through glass	Cockpit windows and drinking glasses look solid	Medium	High
Subsurface scattering for skin	Light under the skin	Faces that look alive rather than plastic	Low	Medium
Subsurface for wax, leaves, ears	Same idea for other materials	Softer, natural look	Low	High
Parallax occlusion mapping	Fake depth in flat surfaces	Carpet, tarmac and brick with real depth	Medium	High
Detail maps	Fine texture repeated close up	Sharp detail when looking closely at seats and desks	Low	Medium
Wear and dirt layers	Scuffs, dust, grime by age	Aircraft and rooms that look used; can follow the aircraft's age in the game	Low	Medium
Wetness	Darker, glossier surfaces in rain	Rain changes the look of runways and aircraft	Low	Medium
Snow and frost cover	Builds up on top surfaces	Winter at Vágar looks like winter; ties into de-icing	Low	Medium
Triplanar mapping	Textures without stretching	Clean terrain and rock textures	Low	All
Material level of detail	Simpler materials at distance	Lower cost with no visible change	Gain	All
Decals	Stickers and marks projected on surfaces	Runway markings, tyre marks, signs, dirt	Low	All
Specular anti-aliasing	Prevents sparkling highlights	Calm, clean shine on metal at a distance	None	All

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7. Textures

Improvement	What it does	Benefit	Cost	Tier
Free scanned material textures (CC0)	Fabric, carpet, leather, wood, metal, concrete, tarmac	Real surfaces instead of flat colours	Low	All
Normal maps	Surface detail without extra shapes	Stitching, panel lines, rivets, grain	Low	All
Roughness and metalness maps	Variation in shine	Surfaces stop looking uniform	Low	All
Higher-resolution textures on Ultra	2K–4K textures up close	Sharpness when walking close to things	Medium (memory)	Ultra

Improvement	What it does	Benefit	Cost	Tier
Anisotropic filtering	Sharp textures at shallow angles	Runways, floors and carpets stay sharp into the distance	Low	All
Procedural textures	Textures created by code	Endless variation with no download size; fits how the game is made	Low	All
Texture variation	Breaks up repeating patterns	No visible tiling on big surfaces	Low	All
Aircraft livery system	Our own liveries with decals and gloss	Each airline's aircraft distinct (own designs only, no real logos)	Low	All
Screen content textures	Real live content on monitors and laptops	Office screens show the game's real data	Low	All
Signs and text rendering (signed-distance fonts)	Crisp text at any size	Sharp signs, gate numbers and labels	Low	All

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8. Anti-aliasing and image quality

Improvement	What it does	Benefit	Cost	Tier
Temporal anti-aliasing (TAA)	Combines frames to smooth edges	Clean edges, no flicker on thin lines (seat rows, runway lights, wires)	Low	Medium
SMAA	Edge smoothing in one frame	Clean edges where TAA is too costly (phones)	Low	All
MSAA tuning	Hardware edge smoothing	Sharp geometry edges	Medium	Medium
Sharpening (contrast-adaptive)	Restores crispness	Clear image after anti-aliasing and upscaling	Low	All
Supersampling on Ultra	Draws above screen resolution	The cleanest image for powerful PCs and screenshots	High	Ultra
Correct colour pipeline	Linear lighting, correct conversion to screen	Natural colours, no washed-out or overly dark scenes	None	All
Modern tone mapping (AgX / Khronos neutral)	Turns bright light into screen colours	Natural highlights and colours, no over-saturated sunsets	None	All
HDR display output	Uses HDR monitors	Brighter lights and deeper contrast on HDR screens	Low	High
Dithering	Prevents colour banding	Smooth skies and gradients	None	All
Wide colour (Display P3)	Richer colours on capable screens	More vivid, accurate colours	None	High

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9. Post-processing and camera effects

Improvement	What it does	Benefit	Cost	Tier
Physically based bloom	Bright lights glow naturally	Runway and cabin lights look bright, not flat	Low	All
Colour grading (LUTs)	A film-style colour look per scene	Distinct mood for offices, court, night flights	None	All
Depth of field	Background blur	Cinematic dialogues; focus on what matters	Medium	Medium
Bokeh	Shaped blur of lights	Beautiful night shots of airport lights	Medium	High
Motion blur	Blur from fast motion	Smoother feel of fast camera moves and take-off rolls	Low–Medium	High

Improvement	What it does	Benefit	Cost	Tier
Lens flare	Light artefacts from the sun	Cinematic sun shots	Low	High
Chromatic aberration	Slight colour fringing	Camera-like realism (subtle)	None	High
Vignette	Darker corners	Focus and mood	None	All
Film grain	Fine noise	Natural, less "computer" look; hides banding	None	Medium
Screen-space reflections	Reflections of what is on screen	Wet runways, glossy floors and windows reflect the world	Medium	High
Heat haze	Air shimmer	Hot exhausts and summer tarmac	Low	High
Rain on the lens / windscreen	Droplets and streaks	Weather you feel in cockpit and outside views	Low	Medium
Frost and condensation on windows	Cold windows	Atmosphere in winter and at altitude	Low	High
Camera exposure and white balance	Like a real camera	Consistent photos of indoor and outdoor scenes	None	All
Outline and selection effects	Clean outlines	Clear highlight of usable objects (alongside the glow)	Low	All

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10. Sky, atmosphere and weather

Improvement	What it does	Benefit	Cost	Tier
Physically based sky and atmosphere	Light scattering through the air	Real blue skies, sunrises, sunsets and twilight	Low	All
Aerial perspective	Distant things fade into the air	True sense of distance and scale from altitude	Low	All
Height fog	Thicker fog in valleys and low areas	Faroese fog in fjords and valleys	Low	All
Volumetric fog	Fog that light passes through	Light beams in fog, lit fog around airport lights	Medium–High	High
Sun shafts (god rays)	Visible beams of sunlight	Beams through cabin windows and clouds	Medium	High
Stars, moon and Milky Way	Accurate night sky	Beautiful night flights	Low	All
Moon light	The moon lights the ground	Night scenes that aren't just black	Low	Medium
Northern lights	Animated aurora	Special winter nights in the north	Low	High
Rain	Falling rain with splashes	Weather matches the simulator's weather	Medium	Medium
Snow	Falling and blowing snow	Winter operations look like winter	Medium	Medium
Lightning	Flashes lighting clouds and ground	Dramatic storms	Low	Medium
Wind-driven effects	Windsocks, grass, flags, spray	The wind the aircraft feels is visible	Low	All
Weather from the simulator	Uses the live weather	What you see matches what you fly	None	All

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11. Clouds

Improvement	What it does	Benefit	Cost	Tier
Volumetric clouds (ray-marched)	3D clouds with real light	Clouds you fly through, lit by the sun	High	Ultra
Low-resolution clouds with upscaling	Clouds drawn at quarter size, rebuilt	Volumetric clouds at a fraction of the cost	Medium	High
Cloud layers from METAR	Cloud bases and amounts from the weather report	Realistic cloud layers at every airport	Low	All
Billboard cloud fields	Many flat cloud images	Good-looking clouds on phones and Low	Low	All
Cloud shadows and light shafts	Clouds shade the ground and let beams through	Living landscapes	Low	High
Cloud flying effects	Moisture, darkening and turbulence inside clouds	Feeling of flying through cloud	Low	Medium
Sunset-lit clouds	Clouds coloured by the sun's angle	Spectacular sunsets and sunrises	Low	All

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12. Water and ocean

Improvement	What it does	Benefit	Cost	Tier
FFT ocean waves	Realistic wave patterns	A real North Atlantic around the Faroes	Medium	High
Gerstner waves	Simpler wave model	Good-looking sea on Medium and phones	Low	All
Wave height from the wind	Sea state follows the weather	Calm or rough seas match conditions	None	All
Foam and whitecaps	Foam on crests and along coasts	Coasts and stormy seas look right	Low	Medium
Shore waves and breaking surf	Waves breaking on cliffs and beaches	Dramatic Faroese coastline	Medium	High
Reflections on water	Sky and terrain reflected	Calm fjords mirror the mountains	Medium	Medium
Sun glitter	Sparkle of sunlight on the sea	Classic flying view over the sea	Low	All
Water colour by depth	Shallow water lighter	Natural coasts and lakes	Low	All
Lakes and rivers	Flat water with ripples	Correct inland water	Low	All

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13. Terrain and ground

Improvement	What it does	Benefit	Cost	Tier
Continuous terrain level of detail (CDLOD)	Terrain detail follows the camera smoothly	Sharp hills up close, wide views, no popping	Low	All

Improvement	What it does	Benefit	Cost	Tier
Terrain materials by slope and height	Grass, rock, scree, snow by terrain shape	Natural mountains instead of only satellite colour	Low	All
Satellite imagery blended with materials	Detail texture added over imagery	Sharp ground up close even with lower-resolution imagery	Low	All
Grass and ground cover (instanced)	Real 3D grass near the camera	Lifelike ground around the airport and on approach	Medium	High
Rocks and boulders	Scattered rocks on slopes	Faroese hillsides look like themselves	Low	Medium
Sheep and animals	Instanced animals in fields	Life in the landscape (fitting for the Faroes)	Low	High
Cliffs and overhangs	Extra geometry where height maps can't show cliffs	Steep sea cliffs look real	Medium	High
Roads and buildings from map data	Towns and roads from OpenStreetMap (ODbL, with credit)	Recognisable places around airports	Medium	Medium
Terrain self-shadowing	Mountains shade valleys	Deep, real landscapes at low sun	Medium	High
Snow cover by season and weather	Snow on high ground	Seasons visible from the air	Low	All

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14. Airports and runways

Improvement	What it does	Benefit	Cost	Tier
Detailed runway surfaces	Tarmac and concrete textures, joints, patches	Runways look real up close	Low	All
Rubber deposits and tyre marks	Darker touchdown zones	Real touch from the air and on landing	Low	All
Correct markings (decals)	Painted markings sharp at any distance	Correct, crisp runway and taxiway markings	Low	All
Edge, centreline and touchdown lights	All runway lighting systems	Night approaches like the real thing	Low	All
Taxiway lights and signs	Lit signs and lights	Night taxi with real guidance	Low	All
Stand guidance and markings	Lead-in lines and stop bars	Parking looks right	Low	All
Terminal and hangar buildings	Generated from map outlines with windows and doors	Each airport has its own buildings	Medium	All
Jet bridges	Moving jet bridges	Real gate operations	Low	Medium
Ground vehicles	Fuel trucks, tugs, buses, catering	Busy, living airports	Low	Medium
Ground crew	Marshallers, loaders (same figure system as everyone)	People at work around your aircraft	Low	Medium
Wet runway reflections	Lights mirrored in wet tarmac	Beautiful rainy night approaches	Medium	High
Snow clearing and de-icing	Ploughed runways, de-icing spray	Winter operations shown	Low	Medium
Fences, poles and antennas	Small airport details	Completes the scene	Low	High

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15. Aircraft exterior

Improvement	What it does	Benefit	Cost	Tier
Panel lines and rivets (normal maps)	Fine surface detail	Aircraft look built, not moulded	Low	All
Moving control surfaces	Ailerons, elevators, rudder, spoilers, flaps follow the simulator	Real aircraft behaviour seen from outside	Low	All
Gear animation and suspension	Gear moves and compresses	Real landings and taxi	Low	All
Engine fans spinning with blur	Fan speed from the simulator	Engines look alive	Low	All
Engine exhaust heat and smoke	Haze and start-up smoke	Realism at start and take-off	Low	High
Navigation, strobe, beacon, logo and landing lights	All exterior lights with correct timing	Real night appearance	Low	All
Wing flex	Wings bend with load	Big jets look right in turbulence and on landing	Low	High
Wingtip vortices and condensation	Vapour in humid air	Spectacular take-offs and landings	Low	High
Contrails	Trails at altitude when conditions allow	Real high-altitude look	Low	Medium
Reflective paint and chrome	Clear coat and reflections	Polished aircraft look	Low	Medium
Weathering by age	Dirt and fading by the aircraft's age in the game	Old aircraft look old; fleet age visible	Low	Medium
Cabin window lights at night	Lit windows from outside	Aircraft look inhabited at night	None	All
Rain and ice on the airframe	Wet and iced surfaces	Weather on the aircraft itself	Low	High

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16. Cockpit

Improvement	What it does	Benefit	Cost	Tier
Live instrument screens	Displays drawn from simulator data	A cockpit that works, not a picture	Low	All
Screen glow and reflections in windows	Displays reflect at night	Real night cockpit	Low	Medium
Backlit panels and flood lights	Panel lighting levels	Night cockpit atmosphere	Low	All
Switch and knob detail	Proper shapes and labels (own designs)	Close-up realism	Low	Medium
Windscreen rain, wipers and ice	Weather on the glass	Approaches in bad weather feel real	Low	Medium
Sun glare and shades	Bright sun and visors	Real light in the cockpit	Low	High

Improvement	What it does	Benefit	Cost	Tier
Head movement on landing and turbulence	Camera follows forces	Physical feeling	None	All

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17. Cabin

Improvement	What it does	Benefit	Cost	Tier
Baked cabin lighting	Soft ceiling and wall light	Photo-like cabin	None	Build
Seat fabric, leather and stitching	Real materials per class	Business and economy look different and real	Low	All
Carpet patterns (own designs)	Textured aisle carpet	Finished cabin	Low	All
Window light that follows the sun	Sun patches move as the aircraft turns	Living cabin	Low	Medium
Window shades opened and closed	Passengers use the shades	Realism; lighting changes	Low	Medium
Seat-back screens	Live moving map and content	Modern cabin	Low	Medium
Galley detail	Trolleys, ovens, coffee makers	Crew service looks real	Low	Medium
Overhead bin open and close	Animated bins	Boarding and disembarking scenes	Low	Medium
Cabin mood lighting	Colour scenes by flight phase	Like modern cabins	None	All
Cabin shake in turbulence	Objects and people move	You feel the flight	None	All

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18. People

Improvement	What it does	Benefit	Cost	Tier
Skin shading	Subsurface light, pores (normal maps)	Lifelike faces in dialogue	Low	Medium
Eyes with depth	Refraction, wet highlight, eye contact	People feel present	Low	Medium
Hair cards	Strands drawn as textured layers	Real-looking hair instead of solid shapes	Medium	High
Cloth wrinkles and folds	Normal maps and simple cloth movement	Uniforms and clothes look worn	Low	Medium
Facial expressions	Smiles, frowns, blinks (with the mouth movement we have)	Emotion in dialogue	Low	Medium
Eye and head tracking	People look at you or at what they talk about	Natural conversations	None	All
Animation blending	Smooth moves between walking, standing, sitting	No sudden pose changes	None	All
Foot placement (inverse kinematics)	Feet stay on floors and steps	No sliding or floating feet	Low	Medium

Improvement	What it does	Benefit	Cost	Tier
Hand placement	Hands hold cups, trays, handles properly	Service and office actions look real	Low	Medium
Crowd variety	Many looks from one figure system	Every passenger different	None	All
Crowd level of detail	Simpler far figures	Full cabins and terminals at good speed	Gain	All

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19. Offices and courtroom

Improvement	What it does	Benefit	Cost	Tier
Baked interior lighting	Soft bounced light	Photo-like rooms	None	Build
Window views	Live outdoor view with the real time of day	Rooms feel part of the world	Low	All
Glass, monitor and desk reflections	Reflection probes	Glossy surfaces look right	Low	All
Wood, leather, fabric and paper materials	Scanned textures	Rich, real materials	Low	All
Small props	Pens, mugs, plants, cables, papers	Rooms feel used and lived in	Low	Medium
Dust in the light	Particles in sunbeams	Atmosphere	Low	High
Live screens	The office monitors show the game's real data	Everything in the room works	Low	All
Ambient occlusion and contact shadows	Darkening where things meet	Furniture stands on the floor, not on it	Low	Medium

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20. Night and lights

Improvement	What it does	Benefit	Cost	Tier
Real light intensities at night	Correct brightness of every light	Night scenes like photos	None	All
Light halos and glare	Glow around bright lights in the air	Lights look bright, especially in fog and rain	Low	All
Town lights from the air	Lights from map data	Night flights over lit towns	Low	All
Lit windows in buildings	Random lit windows	Living towns and terminals at night	None	All
Light pools on the ground	Floodlights light the apron	Real night aprons	Low	Medium
Approach light sequence	Rabbit strobes and sequences	Real night approach (we have the lights; sequencing improves them)	None	All
Night adaptation	Darker but readable scenes	Night is dark but you can still see	None	All

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21. Particles and effects

Improvement	What it does	Benefit	Cost	Tier
GPU particles	Particles calculated on the graphics card	Thousands of particles for spray, smoke, snow at low cost	Gain	Medium
Soft particles	Particles fade where they touch surfaces	No hard edges in smoke and fog	Low	All
Lit particles	Particles receive light and shadow	Smoke and steam that belong in the scene	Low	High
Touchdown tyre smoke	Smoke at touchdown from the landing rate	Landings look and feel real	Low	All
Spray from wet runways	Water thrown up by wheels and engines	Wet operations	Low	Medium
Engine start and exhaust	Smoke and heat at start	Realism	Low	Medium
De-icing spray	Fluid spray	Winter operations	Low	Medium
Coffee steam and cabin service details	Small effects	Life in the cabin	None	High
Birds	Flocks around the coast and airport	Living world (puffins and gulls on the Faroes)	Low	Medium

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22. Animation and motion

Improvement	What it does	Benefit	Cost	Tier
Smooth interpolation of simulator data	Positions blended between updates	Aircraft move smoothly at any frame rate	None	All
Physically based camera motion	Camera with weight and inertia	Cinematic, natural camera moves	None	All
Doors, stairs and hatches	Animated aircraft parts	Real boarding and service	Low	All
Ground vehicle movement	Vehicles drive on taxi lanes	Busy airport	Low	Medium
Flags, windsocks and trees in the wind	Movement from wind data	Living scenes	Low	All
Fade-in of new objects	Level of detail and new objects fade (as we do now)	No popping	None	All

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23. Cameras and views

Improvement	What it does	Benefit	Cost	Tier
Cinematic camera paths	Pre-made camera moves for take-off and landing	Movie-like replays	None	All
Tower and spotter views with zoom lens	Long-lens look, compressed depth	Realistic spotter shots	None	All
Camera shake by events	Shake on touchdown, turbulence, engines	Physical feel	None	All

Improvement	What it does	Benefit	Cost	Tier
Replays	Record and play back a flight	Watch your landing again from any camera	Low	All
Picture-in-picture	A second camera in a small window	Watch outside while in the office or cockpit	Medium	High
Smooth transitions between views	Camera blends, not cuts	Polished feel	None	All

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24. Photo mode and cinematics

Improvement	What it does	Benefit	Cost	Tier
Photo mode	Pause, free camera, adjust focus, exposure and filters	Players make and share beautiful pictures	None	All
Path-traced photos	A still rendered with full ray tracing (open-source path tracer)	Near-film-quality pictures	Seconds per picture	All (PC)
High-resolution screenshots	Pictures larger than the screen	Wallpapers and marketing images	None	All
Cinematic letterbox and grading	Film looks	Trailers and replays	None	All
Video recording of replays	Record in the browser	Sharing flights	Low	High

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25. Settings, scaling and hardware

Improvement	What it does	Benefit	Cost	Tier
Automatic quality choice	Picks Low to Ultra from the graphics card	Good look out of the box on every PC	None	All
Per-effect settings	Each effect switchable	Players fine-tune look against speed	None	All
Frame-rate target	Target 30, 60, 120 fps; quality adjusts	Steady performance	None	All
Battery and heat saving (phones)	Lower settings when hot or on battery	Longer play on phones	Gain	All
Performance overlay	Frame time, draws, memory on screen	Players and testing see what costs what	None	All
Benchmark scene	A fixed flight to compare settings	Players choose settings with real numbers	None	All
Separate settings per view	Different quality for office, cabin and 3D view	Best look where it matters	None	All

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26. Phones and tablets (Android)

Improvement	What it does	Benefit	Cost	Tier
Mobile-friendly texture formats (ASTC / ETC2 via KTX2)	Compressed textures phones understand	Less memory, faster, cooler phones	Gain	All
Baked lighting first	Live effects replaced by baked ones	Rich look at low cost	Gain	All
Half-resolution effects	Effects drawn smaller	Effects on phones at all	Gain	All
Lower draw counts	Instancing and merging	Smooth on mid-range phones	Gain	All
Thermal-aware quality	Lowers quality when the phone gets hot	No throttling slowdowns	Gain	All

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27. Tools for making the game

Improvement	What it does	Benefit	Cost	Tier
Automatic screenshot tests	Pictures of every scene after each change	Graphics never break unnoticed	None	Dev
Performance tests in the build	Fails the build if a scene gets too slow	Performance stays good release after release	None	Dev
Lightmap baking in the build	Lighting calculated when the game is built	Baked lighting always up to date	None	Dev
Texture compression in the build	Textures converted automatically	Small downloads, fast loading	None	Dev
Material preview scene	All materials side by side	Faster tuning of the look	None	Dev
In-game graphics debug view	Shows lighting, normals, overdraw, AO alone	Faster finding of problems	None	Dev

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Suggested order

1. **Foundations (speed first):** update three.js, WebGPU with fallback, instancing, compressed textures, culling, performance overlay. This frees the performance the later steps use.
2. **The biggest visible jump:** ambient occlusion, TAA, baked lighting in the offices, court and cabin, real textures, modern tone mapping, bloom and colour grading.
3. **The world outside:** physical sky and atmosphere, cascaded and soft shadows, terrain materials, runway detail, night lighting.
4. **People:** skin shading, eyes, hair cards, expressions, foot and hand placement.
5. **Weather and water:** rain, snow, fog, ocean, clouds.
6. **Ultra and extras:** volumetric clouds and light, screen-space global illumination, photo mode with path tracing.

Each step comes with before-and-after screenshots and frame-time measurements, and phones get the lighter versions.

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